

Closed Forms and Functional-Form Approximations for $Q_k^{\text{bag},\infty}$

The infinite-bagged excess risk is governed by the *conditional* harmonic mean

$$g_i(k) = k \mathbb{E}_S \left[\frac{1}{A_S} \middle| i \in S \right], \quad R_k^{\text{bag},\infty} = \frac{1}{m^2} \sum_{i=1}^m \tau_i \left(g_i(k) - \frac{1}{\bar{\tau}} \right)^2. \quad (1)$$

Unlike the unconditional $\mathbb{E}_S[1/A_S]$ that yields Q_k^{sub} , the conditioning $i \in S$ blocks a general closed form. It does not, however, block the following.

1. Exact: homoskedastic idiosyncratic block (equicorrelation)

Let the common-loading covariance specialise to a homoskedastic idiosyncratic block,

$$\Sigma_e = \sigma^2 [(1 - \rho) I_m + \rho \mathbf{u} \mathbf{u}'] = D + q \mathbf{u} \mathbf{u}', \quad D = \sigma^2 (1 - \rho) I_m, \quad q = \sigma^2 \rho. \quad (2)$$

Proposition 1 (Exact bagged risk under equicorrelation). *Under (2), every size- k subset is exchangeable, so $w_S = \mathbf{u}/k$ and*

$$Q(S) = \sigma^2 \left[\rho + \frac{1 - \rho}{k} \right] \quad (\text{independent of } S), \quad (3)$$

hence

$$Q_k^{\text{sub}} = Q_k^{\text{bag},\infty} = \sigma^2 \left[\rho + \frac{1 - \rho}{k} \right], \quad \boxed{R_k^{\text{bag},\infty} = \sigma^2 (1 - \rho) z_k} \implies \alpha = 1, \quad (4)$$

with $z_k = \frac{1}{k} - \frac{1}{m}$ and $Q^{\text{opt}} = \sigma^2 [\rho + (1 - \rho)/m]$. No Monte Carlo is required.

Implication. All analytical difficulty, and all of $\alpha > 1$, arises solely from heteroskedasticity in $\{\tau_i\}$ (i.e. a non-scalar block D).

2. Exact: finite-support precisions

Suppose $\{\tau_i\}$ takes G distinct values on groups of sizes $m_1, \dots, m_G$. Then A_S depends only on the group-count vector, which is multivariate hypergeometric, so $g_i(k)$ is a finite sum. For the two-group case (τ_H on m_H forecasters, τ_L on the remaining $m - m_H$):

$$g_H(k) = k \sum_{n=0}^{k-1} \frac{\binom{m_H - 1}{n} \binom{m - m_H}{k - 1 - n}}{\binom{m - 1}{k - 1}} \cdot \frac{1}{\tau_H + n \tau_H + (k - 1 - n) \tau_L}, \quad (5)$$

and symmetrically for $g_L(k)$ (swap roles, use $m_L = m - m_H$). Substituting into (1) gives $R_k^{\text{bag},\infty}$ exactly. The case $m = 2$ is the trivial sub-anchor. This is the appropriate tool in the dominant-forecaster regime, where the Taylor approximation of §3 loses accuracy.

3. Functional-form approximation: general heteroskedastic D

The leave-one-out sum $A_{S \setminus \{i\}} = \sum_{j \in S, j \neq i} \tau_j$ is the sum of $k - 1$ precisions drawn without replacement from the $m - 1$ values $\{\tau_j\}_{j \neq i}$, with hypergeometric mean and variance

$$\mu_i = (k - 1) \bar{\tau}_{-i}, \quad V_i = (k - 1) \frac{m - k}{(m - 1)(m - 2)} \sum_{j \neq i} (\tau_j - \bar{\tau}_{-i})^2, \quad \bar{\tau}_{-i} = \frac{A_m - \tau_i}{m - 1}. \quad (6)$$

Proposition 2 (Second-order leave-one-out expansion). *A second-order expansion of $1/(\tau_i + A_{S \setminus \{i\}})$ about $A_{S \setminus \{i\}} = \mu_i$ gives*

$$g_i(k) \approx \frac{k}{\tau_i + \mu_i} \left[1 + \frac{V_i}{(\tau_i + \mu_i)^2} \right], \quad R_k^{\text{bag}, \infty} \approx \frac{1}{m^2} \sum_{i=1}^m \tau_i \left(g_i(k) - \frac{1}{\bar{\tau}} \right)^2. \quad (7)$$

Small-heterogeneity limit. Writing $\tau_i = \bar{\tau}(1 + \delta_i)$ with $\sum_i \delta_i = 0$ and $V_\delta = \frac{1}{m} \sum_i \delta_i^2$, equation (7) collapses to

$$R_k^{\text{bag}, \infty} \approx \frac{V_\delta}{m \bar{\tau}} \frac{1}{k^2} = \frac{V_\delta}{m \bar{\tau}} z_k^2 (1 + o(1)), \quad \alpha = 2, \quad (8)$$

recovering the Hájek-projection exponent with an explicit constant. The first-order term alone ($V_i \equiv 0$) is exact only at $k = 1$ and understates the decay; the variance term V_i supplies the curvature that sets α .

4. Continuous closed form: Gamma precisions

Modelling $\tau_j \stackrel{iid}{\sim} \text{Gamma}(a, b)$ (rate b , mean $\bar{\tau} = a/b$) and Poissonising gives $A_{S \setminus \{i\}} \sim \text{Gamma}(\nu, b)$ with $\nu = (k - 1)a$, so

$$g_i(k) = k \mathbb{E} \left[\frac{1}{\tau_i + A_{S \setminus \{i\}}} \right] = k b^\nu \tau_i^{\nu-1} U(\nu, \nu, b\tau_i), \quad \nu = (k - 1)a, \quad (9)$$

where $U(\cdot, \cdot, \cdot)$ is the Tricomi confluent hypergeometric function. This is a genuine functional form; it differs from the fixed- $\{\tau_i\}$ value because it additionally integrates over τ_i .

Numerical validation of Proposition 2

Effective exponent α (log-log fit of $R_k^{\text{bag}, \infty}$ on z_k) and median level ratio, exact all-subsets enumeration vs. (7):

profile	α (exact)	α (7)	median ratio
mild heterogeneity	1.99	1.99	1.00
lognormal(0.5)	2.04	2.04	0.99
two-group	1.84	1.86	0.91
strong lognormal(1.0)	1.59	1.74	0.79
dominant (10 vs 1)	1.25	1.49	0.86

Accuracy is essentially exact for moderate heterogeneity and degrades only in the heavy-tail/dominant regime, where the exact finite-support form (5) should be used instead. The ceiling $\alpha = 2$ holds for diagonal D ; correlation structure in Σ_e (factor, Toeplitz) can push the fitted exponent above 2.

5. Approximations from existing theory

The building block $m \bar{v}_{k,i} = \tau_i g_i(k)$ is a simple-random-sample (without replacement) average of a reciprocal. Four established literatures give handles on it.

5.1 Negative-moment (Laplace) identity: exact integral form

Since $A_S > 0$, the elementary identity $\mathbb{E}[1/X] = \int_0^\infty \mathbb{E}[e^{-tX}] dt$ converts the reciprocal into a moment generating function. Over uniform size- k subsets this is an *elementary symmetric polynomial* e_k in $x_j = e^{-t\tau_j}$:

$$\mathbb{E}_S[1/A_S] = \binom{m}{k}^{-1} \int_0^\infty e_k(e^{-t\tau_1}, \dots, e^{-t\tau_m}) dt, \quad (10)$$

$$g_i(k) = k \binom{m-1}{k-1}^{-1} \int_0^\infty e^{-t\tau_i} e_{k-1}((e^{-t\tau_j})_{j \neq i}) dt. \quad (11)$$

Both are exact. In the i.i.d. (Poissonised) limit, with negative-MGF $\phi(t) = \mathbb{E}[e^{-t\tau}]$, the conditional form factorises:

$$g_i(k) \approx k \int_0^\infty e^{-t\tau_i} \phi(t)^{k-1} dt. \quad (12)$$

For $\tau \sim \text{Gamma}(a, b)$, $\phi(t) = (1+t/b)^{-a}$ and (12) integrates to the Tricomi form $g_i(k) = k b U(1, 2-\nu, b\tau_i) = k b^\nu \tau_i^{\nu-1} U(\nu, \nu, b\tau_i)$, $\nu = (k-1)a$, recovering eq. (9) as a special case.

5.2 Finite-population ratio-estimator bias (Cochran)

$g_i(k)$ is the conditional mean of a reciprocal under simple random sampling without replacement; the standard second-order ratio-estimator bias expansion (Cochran, *Sampling Techniques*, 1977) reproduces Proposition 2 exactly, with moments (6) and a remainder

$$\left| g_i(k) - \frac{k}{\tau_i + \mu_i} [1 + V_i/(\tau_i + \mu_i)^2] \right| = O\left(\frac{k \mathbb{E}|A_{S \setminus \{i\}} - \mu_i|^3}{(\tau_i + \mu_i)^4} \right). \quad (13)$$

5.3 U-statistic / Hoeffding–Hájek decomposition and bagging theory

$m \bar{v}_{k,i}$ is a U-statistic of degree k with kernel τ_i/A_S (Hoeffding 1948). Its first-order (Hájek) projection yields the leading term

$$R_k^{\text{bag}, \infty} = \underbrace{\frac{V_\delta}{m \bar{\tau}} z_k^2}_{\text{Hájek projection, } \alpha=2} + \sum_{c \geq 2} \zeta_c(k), \quad V_\delta = \frac{1}{m} \sum_i \delta_i^2, \quad \tau_i = \bar{\tau}(1 + \delta_i), \quad (14)$$

where $\zeta_c(k)$ are the higher Hoeffding ANOVA variance components; these bend the effective exponent away from 2 under strong heterogeneity. This is the formal basis of “ $\alpha = 2$ is the diagonal ceiling” (Bühlmann & Yu 2002, *Ann. Statist.*; Buja & Stuetzle 2006; Friedman & Hall 2007).

5.4 Concentration under sampling without replacement

Bernstein-type inequalities for SRSWOR (Serfling 1974; Bardenet & Maillard 2015) bound $\Pr(|A_S - k\bar{\tau}| > s) \leq 2 \exp\{-s^2/(2k\sigma_\tau^2 + \frac{2}{3}s\tau_{\max})\}$, which controls the Taylor remainder and upgrades the expansions of §3 and §5.2 from approximations to bounds with explicit error.

5.5 Citable anchors

The equicorrelation result (4) is the textbook diversification / global minimum-variance portfolio formula (cf. Kan & Zhou 2007 for the estimated- Σ_e analogue), and the finite- G gap $G^{-1}(Q_k^{\text{sub}} - Q_k^{\text{bag},\infty})$ is exactly the variance of Shapley/coalition sampling at coalition size k (Covert & Lee 2021).

6. The weight-deviation view: dropout, RandNLA, and shrinkage

Adopt the weight-deviation definition

$$R_k^{\text{bag},\infty} = (\bar{v}_k - w^*)' \Sigma_e (\bar{v}_k - w^*) = \sum_{i=1}^m \tau_i^{-1} (\bar{v}_{k,i} - w_i^*)^2, \quad w_i^* = \frac{\tau_i}{m\bar{\tau}}. \quad (15)$$

Then $R_k^{\text{bag},\infty}$ is the squared bias of a *randomly sparsified* estimator of the oracle w^* . The bagged weight is a coordinate-wise shrinkage of w^* ,

$$\bar{v}_{k,i} = \bar{\tau} g_i(k) w_i^*, \quad \bar{v}_1 = \frac{1}{m} \boldsymbol{\iota} \text{ (uniform)}, \quad \bar{v}_m = w^* \text{ (oracle)}, \quad (16)$$

so $\bar{v}_k$ interpolates uniform $\rightarrow$ oracle as $k : 1 \rightarrow m$, and (15) is the squared shrinkage bias. Three literatures approximate this object directly.

6.1 Dropout as adaptive regularization

RSM is structured dropout on the combination weights (keep k of m). In the Bernoulli-keep relaxation (keep probability $p = k/m$ independently per coordinate; this concentrates at fixed size k as $m \rightarrow \infty$), the expected dropout solution is a ridge-shrunk oracle (Wager, Wang & Liang 2013; Baldi & Sadowski 2013):

$$\bar{v}_k \approx \frac{(\Sigma_e + \Lambda)^{-1} \boldsymbol{\iota}}{\boldsymbol{\iota}'(\Sigma_e + \Lambda)^{-1} \boldsymbol{\iota}}, \quad \Lambda = \frac{1-p}{p} \text{diag}(\Sigma_e). \quad (17)$$

Under the budget constraint $\boldsymbol{\iota}' w = 1$ the isotropic part of Λ cancels, so the bias is driven solely by the heterogeneity of $\text{diag}(\Sigma_e)$, reproducing $R_k^{\text{bag},\infty} = O((1-p)^2)$, i.e. $\alpha = 2$, and the rule “homoskedastic $D \Rightarrow$ no bias.” This supplies the regularization interpretation but no predictive content beyond §3.

6.2 Randomized linear algebra / leverage-score sampling

w_S is the constrained solution on a random size- k column subset, and $R_k^{\text{bag},\infty}$ is the expected squared reconstruction error. Randomized-NLA theory (Drineas & Mahoney 2006; Boutsidis, Mahoney & Drineas 2009) shows uniform sampling is variance-suboptimal: the minimizing importance distribution draws coordinate i with probability proportional to its leverage, here $p_i \propto \tau_i$. The penalty paid by *uniform* (flat-prior) RSM is exactly the precision dispersion

$$V_\delta = \frac{\text{Var}(\tau)}{\bar{\tau}^2} = \frac{1}{m} \sum_i \delta_i^2, \quad R_k^{\text{bag},\infty} \approx \frac{V_\delta}{m\bar{\tau}} z_k^2, \quad (18)$$

matching (8)/(14). This motivates an *importance-sampled RSM* ($p_i \propto \tau_i$) with strictly smaller $R_k^{\text{bag},\infty}$, and shows the flat subset prior of §(BMA comparison) is not the variance-optimal sampler.

6.3 Shrinkage estimators and a spectral bound

By (16), $\bar{v}_k$ is a coordinate-heterogeneous shrinkage of w^* toward the uniform vector $\frac{1}{m}\mathbf{1}$; linear-shrinkage MSE results (Ledoit & Wolf; James–Stein) provide an envelope on $R_k^{\text{bag},\infty}$ (loose, as RSM’s shrinkage is non-scalar). Finally, since $\bar{v}_k - w^*$ lives in the span of Σ_e , $R_k^{\text{bag},\infty}$ is bounded by the tail eigenvalue mass of Σ_e ,

$$R_k^{\text{bag},\infty} \leq \lambda_{\max}(\Sigma_e) \|\bar{v}_k - w^*\|_2^2, \quad (19)$$

linking the geometric decay observed for the Toeplitz DGP to the Σ_e spectrum.

6.4 What leverage sampling pins down for *uniform* RSM

The leverage framework does more than flag suboptimality; it *derives* $R_k^{\text{bag},\infty}$ for our case via two identifications.

(i) Oracle weight = leverage score. In the common-loading model the constrained-GLS influence of forecaster i is

$$\ell_i = \frac{\tau_i}{A_m} = w_i^*, \quad (20)$$

so the statistical leverage scores are exactly the oracle weights, and the variance-optimal RandNLA proposal $p_i \propto \ell_i$ is “sample in proportion to w^* .”

(ii) Uniform RSM is self-normalized importance sampling. A single size- k draw estimates w_i^* by the self-normalized ratio τ_i/A_S (unnormalized weight τ_i , normalized over the drawn set S); infinite bagging Rao–Blackwellizes it,

$$\bar{v}_{k,i} = \mathbb{E}_S \left[\frac{\tau_i \mathbf{1}_{i \in S}}{A_S} \right], \quad (21)$$

which is the Rao–Blackwellized self-normalized (SNIS) estimator of the leverage vector under a *uniform* proposal. The SNIS bias equals the ratio-estimator bias of §5.2; thus leverage theory *derives* $R_k^{\text{bag},\infty}$ but does not improve on §3.

(iii) Sandwich bound and the role of coherence. Combining the bagging- $\leq$ -subbagging inequality with the Taylor expansion of $\mathbb{E}_S[1/A_S]$ (§4.1 of the paper) and the diagonal Hájek ceiling (14):

$$\underbrace{\frac{V_\delta}{m\bar{\tau}} z_k^2 (1 + o(1))}_{\text{diagonal Hájek, } \alpha=2} \lesssim R_k^{\text{bag},\infty} \leq Q_k^{\text{sub}} - Q^{\text{opt}} \leq \frac{z_k}{\bar{\tau}} (1 + o(1)) \implies 1 \leq \alpha \leq 2. \quad (22)$$

Where α falls in $[1, 2]$ is governed by the *precision coherence*

$$\kappa = \frac{\tau_{\max}}{\bar{\tau}} = m \max_i \ell_i : \quad (23)$$

low coherence $\Rightarrow$ uniform $\approx$ leverage sampling $\Rightarrow \alpha \rightarrow 2$; high coherence (a dominant forecaster) $\Rightarrow$ uniform is far from the variance-optimal proposal $\Rightarrow \alpha \rightarrow 1$. This matches the empirical dominant-forecaster regime. The left bound in (22) is the leading-order value under mild heterogeneity, not a uniform lower bound; the right inequalities are exact.

Numerical validation of (23). Effective exponent α (log-log fit of the exact all-subsets $R_k^{\text{bag}, \infty}$ on z_k , $m = 9$) against the precision coherence $\kappa = \tau_{\max}/\bar{\tau}$:

precision profile	coherence κ	α
near-homogeneous	1.09	2.00
mild heterogeneity	1.37	2.01
lognormal(0.5)	1.48	2.09
lognormal(1.0)	2.37	1.75
two-group	2.65	1.80
dominant	5.40	1.24
very dominant	7.50	1.08

The map $\kappa \downarrow \Rightarrow \alpha \uparrow$ is monotone across the range: $\kappa \rightarrow 1$ gives $\alpha \rightarrow 2$ (uniform sampling near-optimal), large κ gives $\alpha \rightarrow 1$ (uniform far from the leverage-optimal proposal). Reproduced by `coherence_alpha_check.py`.